A methodological approach to energy optimization for buildings by utilizing machine learning-based surrogate model and an optimization technique

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ABSTRACT

This study presents an application procedure for architectural design of residential buildings in Vietnam, with the objective of reducing environmental effects. Particularly, this practical procedure utilizes software to simulate the energy usage of the building and then applies a machine learning model to build an energy surrogate model. The applicability of the surrogate model is guaranteed by its high accuracy, with the relative error of only around 1.5%, as well as by a high correlation of greater than 0.9 between the predictions and true values. An optimization framework is formulated to identify an optimal architectural design by minimizing the building's energy requirements through an objective function based on the surrogate model and subject to architectural constraints aligned with local design principles. The optimization process is executed using the Constriction Particle Swarm Optimization algorithm. The algorithm shows fast convergence, which is within 40 iterations. The presented procedure is illustrated through a case study of a residential building located in An Giang province, Vietnam, utilizing the most recent climate data officially published by the Vietnamese Government. The results demonstrate the methodology's potential for practical design applications, with key discussions provided on the findings and future research directions.

1. Introduction

The world has faced climate issues over the past decade, such as a rise in the Earth's surface temperature of 1.1 degrees Celsius (Intergovernmental Panel on Climate Change, 2023). The provisional State of the Global Climate reported by the World Meteorological Organization (WMO) (World Meteorological Organization (WMO), 2023) indicates that, as of the end of October 2023, the year's temperature was approximately 1.4 degrees Celsius above the pre-industrial baseline from 1850 to 1900.

Vietnam is one of the regions that is significantly impacted by climate change. According to the Climate Risk Country Profiles from the World Bank Group and the Asian Development Bank. Since 1960, the mean annual temperature in Vietnam has risen by 0.5 to

0.7 degrees Celsius, with the most rapid increases occurring in southern Vietnam and the Central Highlands. Currently, the daily likelihood of a heat wave is roughly 3% (World Bank, 2019). Future climate projections indicate that by the end of the century, temperatures could rise by approximately 1.3 degrees Celsius under a low greenhouse gas emissions scenario (RCP2.6) and about 4.2 degrees Celsius under a high emissions scenario (RCP8.5) (World Bank, 2019).

Buildings account for over 40% of global energy consumption and approximately one-third of greenhouse gas emissions (International Energy Agency & United Nations Environment Programme, 2018; United Nations Environment Programme, 2009). The building industry places a significant strain on the environment due to emissions generated from producing building materials and operating building systems. To address these impacts, various initiatives have focused on designing and retrofitting buildings for energy efficiency and sustainability. Numerous studies have explored ways to reduce energy demand and carbon emissions in buildings. One review (Mostafavi et al., 2021) analyzed 48 studies on the energy and carbon performance of high-rise buildings from 2005 to 2020 across different climatic conditions. The findings indicated substantial potential for reducing energy consumption in high-rises through improvements in envelope design (up to 78.9%), optimizing floor plans (up to 17%), and implementing natural ventilation strategies (up to 45%). A study conducted in China (Hu & Zheng, 2015) evaluated energy consumption and carbon emissions using a life-cycle carbon emission calculation methodology for residential buildings. Carbon emissions were assessed through Life Cycle Assessment (LCA). The study analyzed various influencing factors, including insulation thickness, types of air conditioning, and the buildings' service life. Based on the analysis results, the researchers identified several strategies for reducing emissions and outlined their potential for impact. A case study conducted on a typical office building in Thailand revealed that steel and concrete are the most significant materials used, both in terms of the quantities consumed and their associated environmental impacts during the manufacturing process. The research highlighted that a substantial 42% of the life cycle carbon emissions are attributable to the embodied carbon footprint linked to the production of these construction materials. This means that the environmental impact of a building begins long before it is occupied, primarily due to the energy-intensive processes required to produce steel and concrete (Kofoworola & Gheewala, 2008).

Given the significant impact of climate change on Vietnam and the role of the building sector in exacerbating environmental challenges, it is essential to address the industry's environmental footprint. A key strategy is to minimize the energy consumption of buildings, which directly contributes to reducing greenhouse gas emissions and mitigating climate change.

In recent years, Machine Learning (ML) has seen significant growth across various engineering applications. For instance, ML has been extensively used to predict the mechanical properties of concrete (Chaabene et al., 2020), as well as in materials science (Cao et al., 2024), and environmental science (Zhong et al., 2021). One of the most significant advantages of ML-based methods is its ability to reduce the need for complicated mathematical models required within traditional computation frameworks, thereby offering great flexibility and potential for solving challenging problems. These advancements have led to a substantial change in how engineering problems are addressed, moving the emphasis from traditional mathematical models to data-driven ML models. There has been a wide range of fields which utilize ML-based surrogate modeling for solving problems, such as urban water network (Garzón et al., 2022), groundwater contaminant modeling (Luo et al., 2023), aerodynamic design (Sun & Wang, 2019), and chemical process engineering (McBride & Sundmacher, 2019).

Over the past five years, there has been significant progress in using ML to evaluate and optimize building energy performance. Recently, Boutahri and Tilioua (2024) emphasized that efficient energy use and occupant comfort are key considerations in building design and operation. They introduced an ML-based model to predict thermal comfort levels and optimize energy consumption in Heating, Ventilation, and Air Conditioning (HVAC) systems, utilizing four ML methods: Support Vector Machine, Artificial Neural Network, Random Forest, and Extreme Gradient Boosting. Egwim et al. (2024) conducted a comparative study to assist in the initial selection of appropriate ML algorithms for building energy analysis, evaluating widely used models for assessing energy efficiency.

Olu-Ajayi et al. (2022) focused on improving energy efficiency during the design stage of buildings. Their study compared nine classification-based machine learning algorithms for assessing the energy performance of residential buildings, incorporating feature selection and hyperparameter tuning. Wu et al. (2022) proposed an intelligent optimization framework using ML to achieve near-zero energy consumption while balancing building comfort and cost. This framework integrates Building Information Modeling (BIM) with DesignBuilder and the Random Forest-Nondominated Sorting Genetic Algorithm III. Yang et al. (2020) developed a model predictive control system based on adaptive machine learning, using real-time building operation data to update the model. This system optimizes energy efficiency and indoor thermal comfort using a multi-objective approach, focusing on air-conditioning and mechanical ventilation systems for experimental evaluation. Seyedzadeh et al. (2019) reviewed the effectiveness of common ML models in estimating heating and cooling loads for buildings.

García-Martín et al. (2019) underscored the importance of evaluating energy consumption in buildings and reviewed various methodologies with a focus on ML applications. The growing integration of technology in smart buildings has drawn significant attention to ML's potential in construction. Alanne and Sierla (2022) examined the learning capabilities of buildings from a system-level perspective, highlighting the role of ML in building energy management. Additionally, motivated by the fact that buildings in developed countries account for a large share of total energy consumption and contribute significantly to global greenhouse gas emissions, Fathi et al. (2020) explored ML applications for urban-scale building energy performance forecasting from 2015 to 2018. In addition to the recent studies on ML for building energy issues, earlier research examined commercial building energy consumption estimation (Robinson et al., 2017), building energy prediction (Singaravel et al., 2017), and energy use predictions during architectural concept design (Paterson et al., 2017). A comprehensive overview of ML for estimating building energy consumption and performance can be found in paper of Seyedzadeh et al. (2018).

Energy optimization in buildings involves managing both energy consumption and conservation to enhance environmental sustainability and human well-being. For example, in smart buildings, energy optimization often focuses on controlling systems such as lighting, temperature, ventilation, and water heating and cooling to minimize energy use, typically after construction is completed. In this study, we aim to present a procedural method for optimizing the energy consumption of residential buildings in Vietnam, taking into account architectural aspects, based on techniques, approaches which have widely been established and studied within the field. Since a building's architecture remains fixed throughout its lifespan, it is essential to consider these factors during the design phase of construction. When combined with optimized control systems, architectural optimization can significantly enhance overall energy performance.

Simulating energy-related quantities in buildings demands significant computational resources due to the complex mathematical modeling of thermal physics and mechanics. Additionally, solving an optimization problem requires numerous evaluations of the objective function, each of which can be computationally expensive, posing a challenge in developing an optimization solution. A practical approach to this issue is creating a surrogate model that can directly map design variables to the desired output. ML algorithms excel in this area by learning from the data's inherent features without needing to solve complex mathematical equations. As more data becomes available, the predictive power of ML models improves, making them an ideal choice for surrogate modeling. For instance, in the paper published by Prina et al. (2024), researchers combined machine learning algorithms with energy simulation software to develop a surrogate model, which was integrated into a multi-objective evolutionary algorithm to optimize energy performance in complex building models. In this study, we aim to develop a surrogate model using ML algorithms to estimate the building's energy consumption. This model will then be used to calculate the objective function within the optimization framework.

This study aims to develop a procedural methodology for identifying optimal solutions to enhance building energy efficiency. At this stage, we steer our work to an application-oriented study whose objective is to provide an engineering method for building design with the reduction of environmental effects being considered. First, the general design scheme applicable to a residential building's design is presented. This will be followed by a case study that follows the aforementioned procedure.

Specifically, the study is performed through the following parts. The building's energy performance is assessed through simulations conducted using specialized software. A machine learning-based surrogate model is developed to map the design variables directly to the building's energy consumption, using a database generated from these simulations. This surrogate model serves as the basis for constructing an objective function that minimizes energy requirements while satisfying architectural constraints. The optimization is implemented Constriction Particle Swarm Optimization (CPSO) algorithm. The presented procedure is realized through a case study of a residential building located in An Giang province, Vietnam. The energy evaluation leverages the most recent climate data recorded in 2018 and officially published by the Vietnamese Government. Additionally, architectural constraints aligned with local design principles for residential buildings are incorporated into the optimization process.

The paper is organized as follows: Section 2 outlines the proposed methodology. Section 3 demonstrates its application through a case study. Section 4 discusses key findings and presents future research directions based on the results. Finally, Section 5 provides the conclusion of the study.

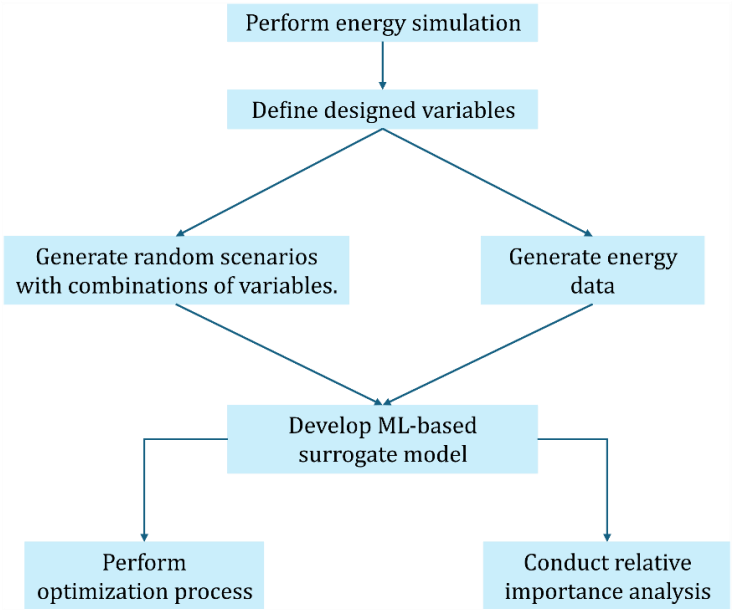
2. Methodology

In this section, we present parts of our presented design procedure, which are arranged in a step-by-step manner. Our approach addresses the problem of minimizing building energy consumption by optimizing variables related to the building's design. First, energy consumption is simulated using TerMus PLUS, a Building Energy Modeling (BEM) software developed by ACCA. The design variables for the problem are then determined based on the building's architectural strategy, with upper and lower bounds assigned to each variable. A random set of variable combinations is generated within these bounds, creating a sample set of input data. Each combination is used as input for BEM simulations to calculate energy consumption. Once all cases have been simulated, a database is obtained, mapping the design variables to their

corresponding energy consumption. An ML-based surrogate model is then developed from this database to approximate the building's energy consumption based on the design variables. This surrogate model is subsequently used to construct the objective function for the optimization process. The procedure is illustrated in the flowchart shown in Figure 1, and the details of each step are described below.

Figure 1

General Flowchart of the Proposed Methodology



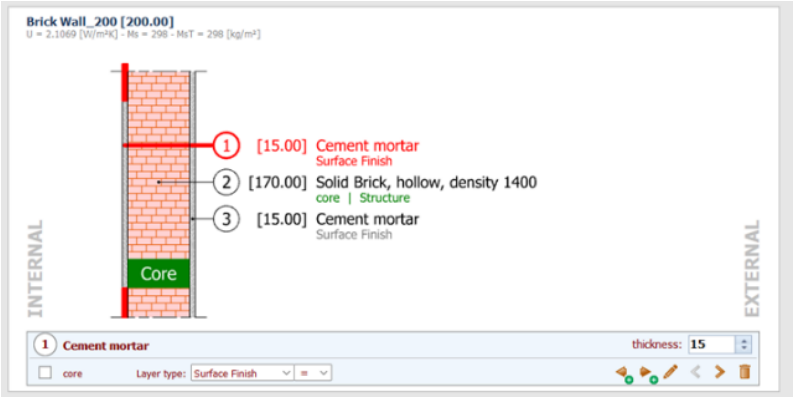
Note. Complied by the authors

2.1. Energy simulation

This part outlines the key features of the energy simulation applied in this study. Initially, the in-plane boundary and height reference system are established. For the calculation of energy performance, BEM-objects are defined with relevant energy parameters. The parameters basically include thermal properties and geometric details. For example, Figure 2 illustrates the definition of a solid brick wall, showing the different materials used across its layers and the geometry of the object.

Figure 2

A Definition of a Solid Brick Wall

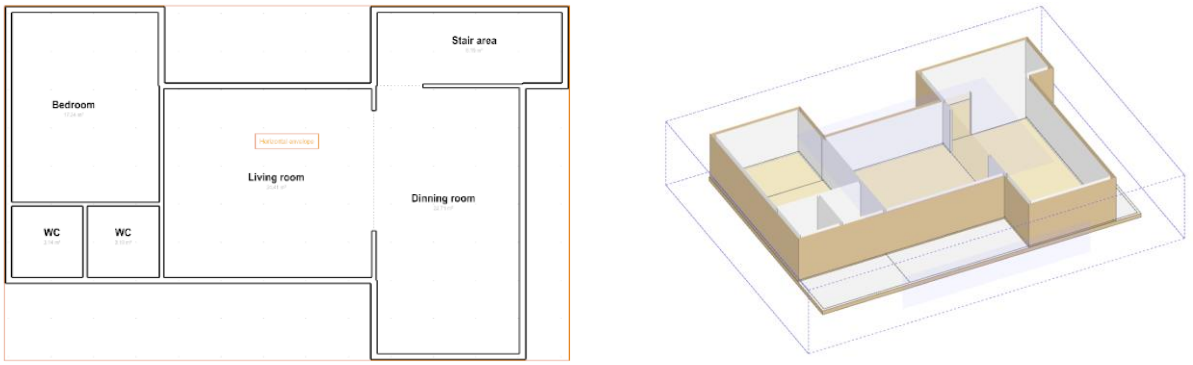


Note. Complied by the authors

The BEM-objects are categorized into two main groups: vertical entities and horizontal entities. The former includes walls, windows, and doors, while the latter consists of slabs, ceilings, and the roof. Window and door entities may contain sub-entities, such as glass and frames. Energy parameters of each entity are defined in this stage. Once all BEM-objects are defined, they are modeled based on the architectural design and the reference system established in the initial step. Figure 3 depicts a stage in the modeling process of the wall elements on the building's first floor.

Figure 3

Wall Modeling for First Floor



(a) Floor plan (b) 3D rendering

Note. Complied by the authors

In building energy simulation, it is crucial to account for the thermal behavior at intersections of elements with differing thermal properties, a phenomenon known as a "thermal bridge" in energy modeling. This approach addresses the discontinuity in thermal properties at the intersection by modeling it as a thermal bridge object with an equivalent thermal property. The locations where thermal bridges occur are first identified, and the thermal bridges are then calculated using TerMus PLUS software.

Once the energy model for the building is established, energy boundaries and thermal data are defined. In detail, an energy boundary refers to an energy zone assigned to a specific space within the building, based on the intended function of that space. The building's energy zones are categorized into four types: heating, cooling, domestic hot water, and ventilation zones. Depending on the use of a particular area, one or more relevant energy zones are assigned to it, allowing their contribution to the building's overall energy consumption to be accurately assessed. After defining energy zones for the spatial areas, a usage schedule for each zone is specified for the entire year. This enables the calculation of the building's energy consumption, with results presented in daily, monthly, and yearly summaries.

Subsequently, the properties of the HVAC and DHW systems, which serve the previously defined energy zones, are configured. The operation schedules of these energy zones are incorporated into the settings of the HVAC and DHW systems. Based on these schedules and the specified energy properties of the systems, the energy usage of these systems is calculated. It is important to note that multiple energy zones may be assigned to a single area; for instance, a living room may be designated with both cooling and ventilation modes.

Next, the external weather conditions are declared in the software, through which the difference between indoor and outdoor temperatures can be measured. The empirical data of external weather conditions most recently recorded is used for the declaration.

Finally, after all steps are completed, the energy analysis of the building can be performed. The simulation results include energy data for both the building's functional zones and the HVAC and DHW systems. For the functional zones, the results provide information on energy requirements, energy losses, and heat gains from external environments. For the HVAC and DHW systems, the analysis outputs the energy requirements, operational costs, and associated carbon emissions.

2.2. Designed variables

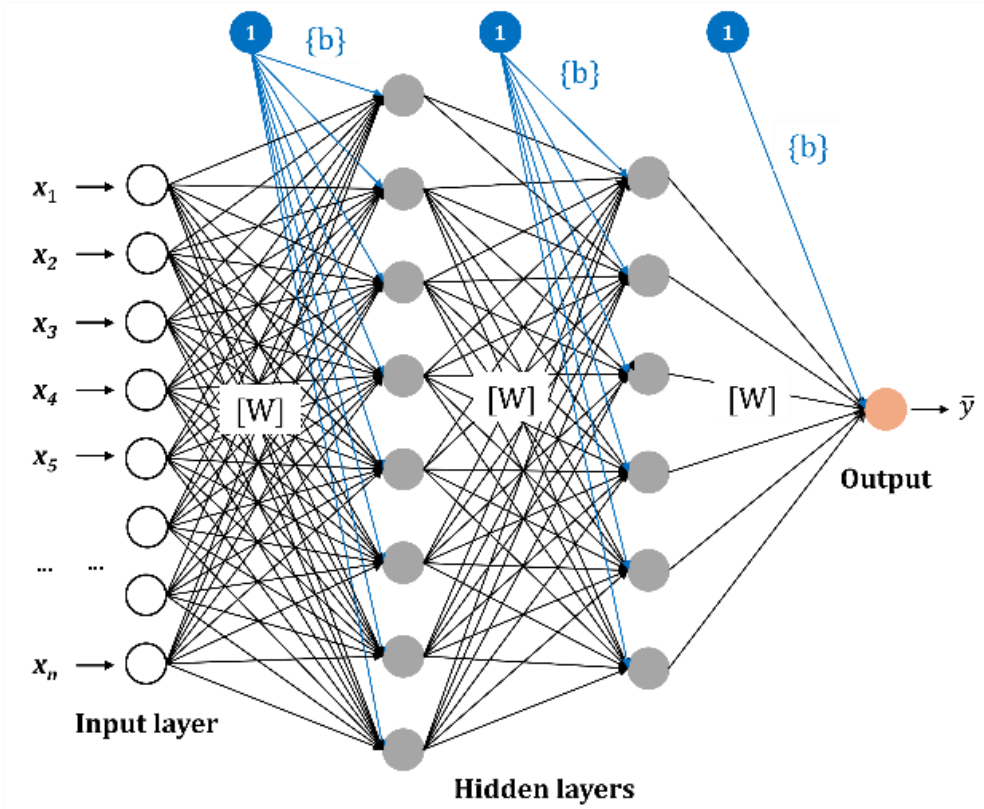
Based on the architectural design strategy, the variables for the optimization process are selected. For example, energy performance varies depending on changes in architectural factors such as window area, wall thickness, and door geometry. Next, to establish a database for building a surrogate model, a random sampling process based on those is performed on this set of variables. Afterward, energy simulations are conducted for each random scenario, generating a database that captures the relationship between architectural factors and energy consumption. This database is subsequently used to develop an ML-based surrogate model, which is used to develop the objective function for the optimization process.

2.3. ML-based surrogate model

In our work, a Machine Learning (ML) approach utilizing an Artificial Neural Network (ANN) is employed to develop a surrogate model, trained with the database generated from the previous step. A typical ANN model consists of three main components: an input layer, one or more hidden layers, and an output layer, as illustrated in Figure 4.

Figure 4

Structure of a Typical ANN Model



Note. Complied by the authors

In an ANN model, each node within a hidden layer performs two key operations: a linear summation and an activation transformation, as expressed in the following equation.

$$a_j^{(l)} = \mathcal{A}\left(\sum_{k=1}^{nL^{(l-1)}} a_k^{(l-1)} w_{kj}^{(l)} + b_j^{(l)}\right) \tag{1}$$

where $a_j^{(l)}$ is the output value at j^{th} node of lth hidden layer, $a_k^{(l-1)}$ is the output value at kth node of $(l-1)^{th}$ hidden layer, $w_{kj}^{(l)}$ is the weight factor connecting k^{th} node and j^{th} node, nL is the number of layers, and $\mathcal{A}(\cdot)$ is the chosen activation function.

The surrogate model establishes a direct mapping between the values of the designed variables and the corresponding energy consumption of the building. In this study, standardization is applied to the input variables to ensure they have a mean of 0 and a standard deviation of 1, enhancing the performance and convergence of the machine learning model. Each variable x_i is transformed using the following formula:

$$z_i = \frac{x_i - \mu}{\sigma} \tag{2}$$

where μ is the mean and σ is the standard deviation of the variable. This transformation ensures that all input variables are on a comparable scale, preventing any single variable from disproportionately influencing the optimization process. Additionally, the early stopping technique is employed to prevent overfitting, which can occur during the training of the ML model.

Additionally, to evaluate the performance of the ANN model in predicting energy consumption, the dataset is divided into training and testing subsets. The model's performance is assessed by comparing evaluation metrics calculated for both sets. In this study, the correlation coefficient (R), Mean Squared Error (MSE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE) are employed as evaluation metrics. The formulations of these metrics are provided below.

$$R = \frac{n\sum_{i=1}^{n}\bar{y}_i y_i - (\sum_{i=1}^{n}\bar{y}_i)(\sum_{i=1}^{n} y_i)}{\sqrt{n(\sum_{i=1}^{n}\bar{y}_i^2) - (\sum_{i=1}^{n}\bar{y}_i^2)^2}\sqrt{n(\sum_{i=1}^{n} y_i^2) - (\sum_{i=1}^{n} y_i^2)^2}} \tag{3}$$

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\bar{y}_i - y_i)^2 \tag{4}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\bar{y}_i - y_i)^2} \tag{5}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\bar{y}_i - y_i| \tag{6}$$

where $\bar{y}_i$ and y_i are the predicted and true values of the ith sample, respectively.

2.4. Optimization for the energy problem

Using the developed ANN-based surrogate model developed from the previous step, an objective function which accounts for the energy requirement of the building can be formulated. To ensure a feasible design, architectural constraints for the designed variables should be incorporated into the optimization framework. These constraints are defined according to architectural design principles that must be adhered to for specific building types. For example, a good design principle for residential buildings in some Asian countries requires the ratio of total wall thickness to floor area to be less than 15%.

In this study, the Constriction Particle Swarm Optimization (CPSO) algorithm (Clerc & Kennedy, 2002) is employed to solve the formulated optimization problem. CPSO is a refined version of the Particle Swarm Optimization (PSO) method (Kennedy & Eberhart, 1995) designed to enhance convergence stability. The CPSO algorithm is illustrated in Figure 5, and the velocity and position updates for dimension *d* are governed by the following equations:

Velocity update

$$v_{i,t+1}^{d} = \chi\left(v_{i,t}^{d} + c_1 r_1\left(p_{i,t}^{d} - x_{i,t}^{d}\right) + c_2 r_2\left(g_i^{d} - x_{i,t}^{d}\right)\right)$$

$$\chi = \frac{2}{\phi - 2 + \sqrt{\phi^2 - 4\phi}} \tag{7}$$

$$\phi = c_1 + c_2$$

where $v_{i,t}^{d}$ is the velocity of particle *i* at iteration *t*; $p_{i,t}^{d}$ is the personal best position of particle *i* until the iteration *t*; g_i^{d} is the global best position found in the swarm; c_1 and c_2 are cognitive and social coefficients; r_1 and r_2 are random number between 0 and 1; χ is the constriction factor.

Position update

$$x_{i,t+1}^{d} = x_{i,t}^{d} + v_{i,t+1}^{d} \tag{8}$$

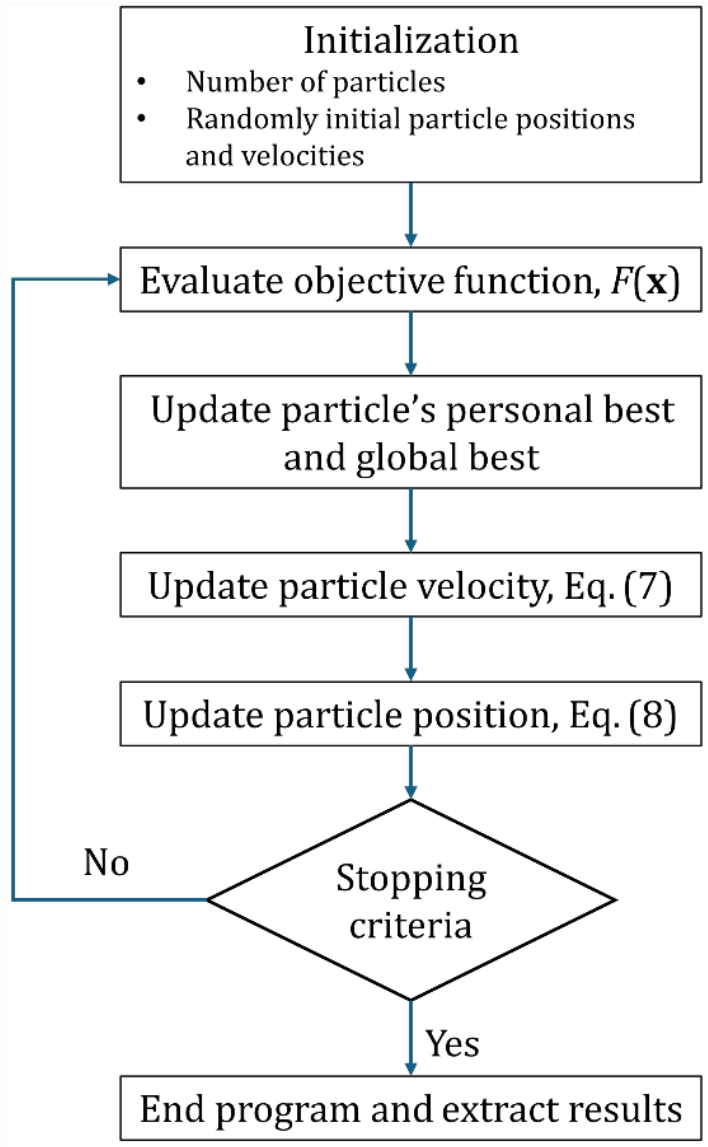
where $x_{i,t}^{d}$ is the position of the particle *i* at iteration *t*.

3. A case study for the proposed approach

In this section, the presented procedure described above is implemented through a case study. This case study focuses on identifying optimal architectural design solutions for residential buildings of a typical type in Vietnam. The selected building is situated in An Giang province, located in the southwest of Vietnam. The implementation is conducted step-by-step based on the above-presented skeleton. First, general information about the building, including its type and local climatic conditions, is provided. Next, the generation of the database is described, followed by the designation of the design variables. Subsequently, the development of the ANN-based surrogate model is presented, which forms the basis for constructing the objective (loss) function in the optimization process. Finally, some discussions are made.

Figure 5

General Flowchart for the CPSO



Note. Complied by the authors

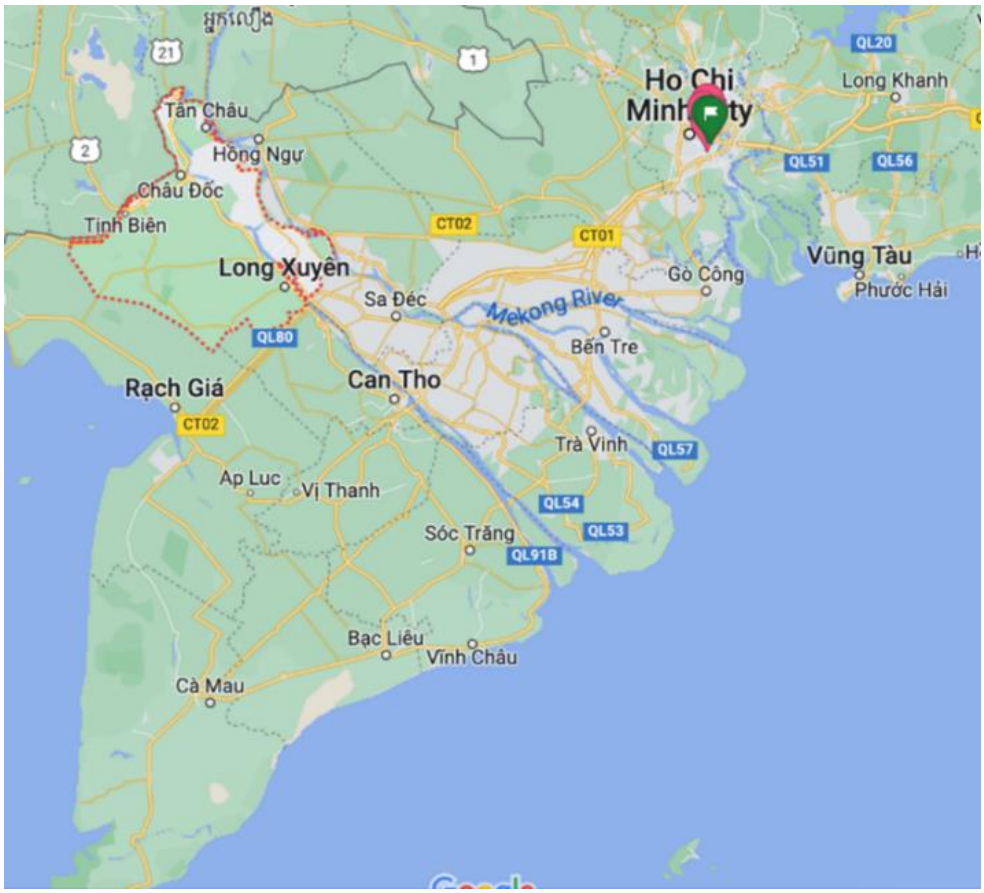
3.1. General information about the building

The building is located in An Giang province, situated in the southwest of Vietnam, bordering Cambodia. Figure 6 shows the geographical location of the building. Overall, the average annual temperature ranges from 26 to 28 degrees Celsius, with peak temperatures of 35 to 36 degrees Celsius occurring in April and May. The lowest temperatures, between 20 and 21 degrees Celsius, are typically observed in December and January.

In terms of a general description of the building, it adopts a townhouse style with a simple design, light-colored finishes, and a sloping roof. The ground floor accommodates functional spaces such as a living room, bedroom, kitchen and dining room, two toilets, staircase, and gardens. The first floor comprises three bedrooms, a living room, two toilets, laundry and drying spaces, and staircase. The architecture of the building is detailed in Figure 7 and 8.

Figure 6

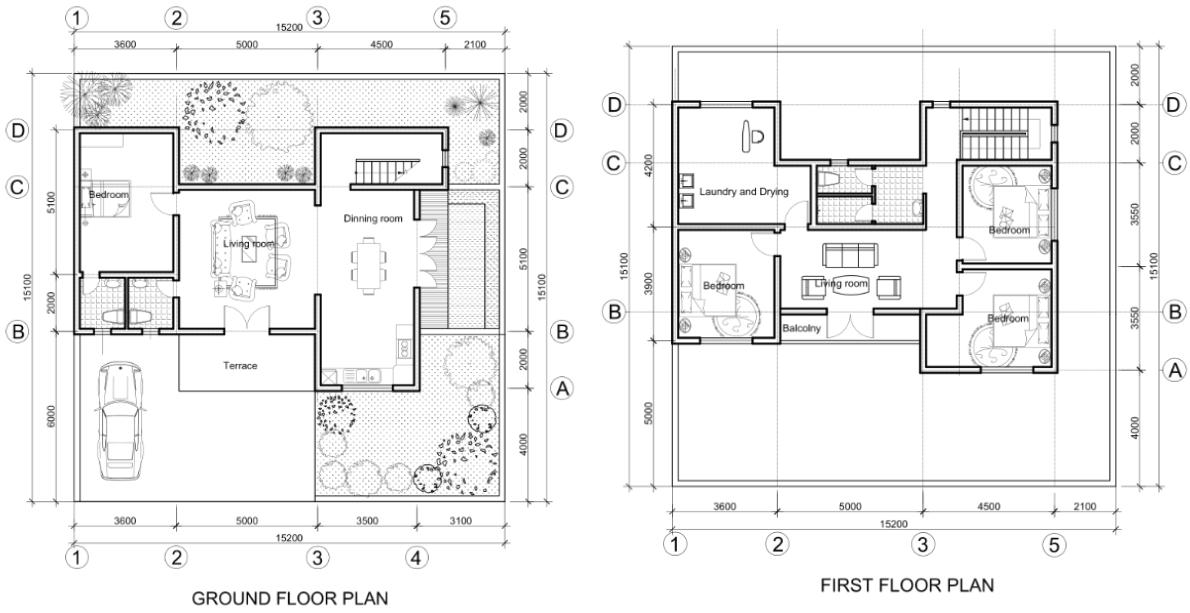
Geographical Location of the Building (maps.google.com)



Note. Complied by the authors

Figure 7

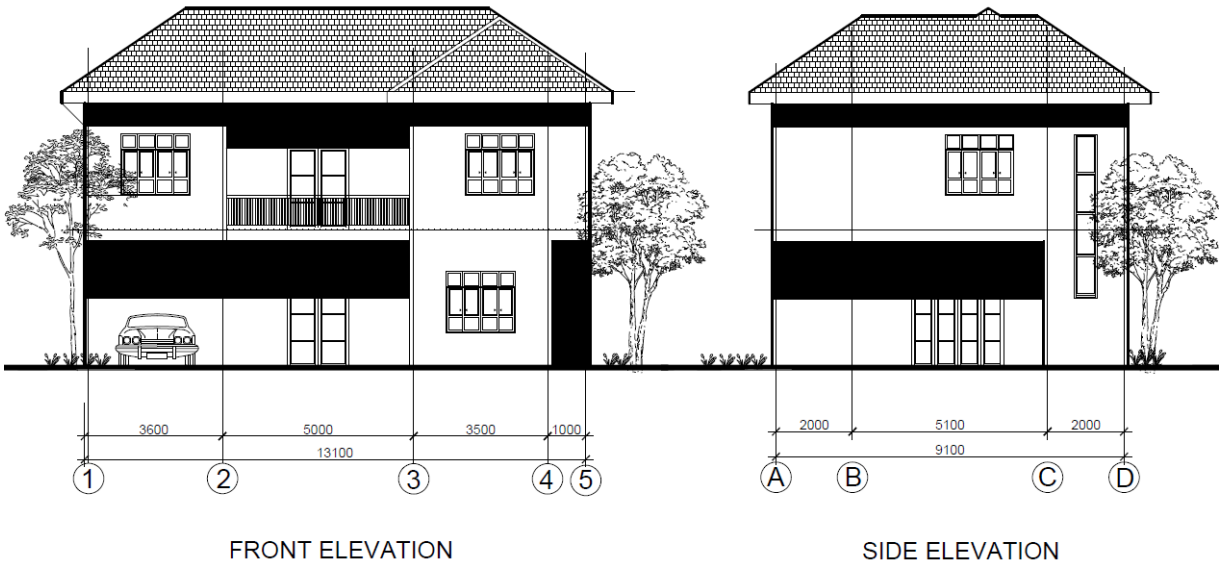
Architectural Plan Views of the Building



Note. Complied by the authors

Figure 8

Architectural Elevation Views of the Building



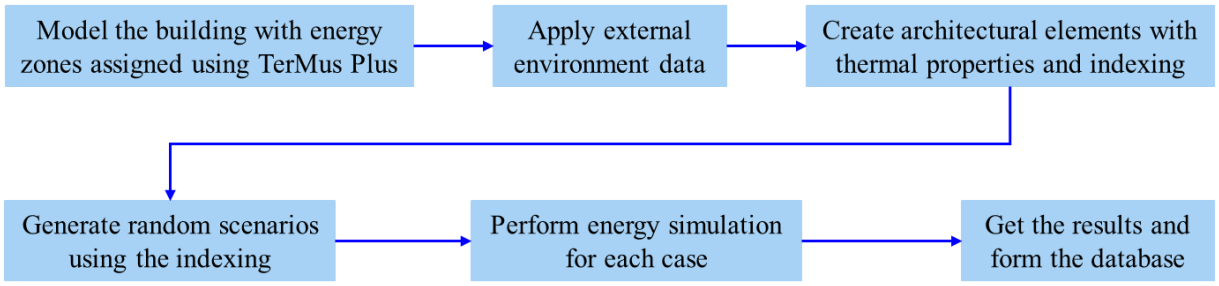
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3.2. Energy simulation

In this case study, eight varying architectural features are considered for the energy simulation, which serves as the foundation for developing the ML-based surrogate model. Specifically, the features include slab thickness, ceiling thickness, external wall thickness, internal wall thickness, external window area, number of external window layers, primary door area, and secondary door area. The procedure of data generation and preprocessing is indicated in Figure 9.

Figure 9

Data Generation Procedure



Note. Complied by the authors

To evaluate the building's energy performance across different configurations, a BEM library is created containing elements corresponding to these architectural features, each with specified parameters. A categorical encoding procedure is applied to all BEM elements in the library. Tables 1 to 6 present the encoded BEM objects prepared for the random combinations of variable scenarios, used for simulations. In those tables, the first column tells the indexing number of each architectural element used for generating the database. The following columns tell the name and the physical properties of the elements. For example, in Table 1, "Concrete slab 130" is the name for a concrete slab element which is indexed by 1 and has thickness of 130 (mm).

Table 1

Slab Library

Slab index	Title	Thickness (mm)
1	Concrete slab 130	130
2	Concrete slab 150	150
3	Concrete slab 200	200

Note. Complied by the authors

Table 2

Wall Library

Wall index	Title	Thickness (mm)
1	Brick wall 100	100
2	Brick wall 150	150
3	Brick wall 200	200
4	Brick wall 250	250

Note. Complied by the authors

Table 3

Window Library

Window index	Title	Area (m^2)	Thickness (mm)
1	S6 1015 2HDL	1.5	1
2	S6 1215 2HDL	1.8	1
3	S6 1415 2HDL	2.1	1
4	D6 1015 2HDL	1.5	2
5	D6 1215 2HDL	1.8	2
6	D6 1415 2HDL	2.1	2

Note. Complied by the authors

Table 4

Primary Door Library

Primary door index	Title	Area (m^2)
1	1622 2HDL	3.52
2	1822 2HDL	3.96
3	2022 2HDL	4.4

Note. Complied by the authors

Table 5

Secondary Door Library

Secondary door index	Title	Area (m^2)
1	0621 1HDL	1.26
2	0821 1HDL	1.68

Note. Complied by the authors

Table 6

Ceiling Library

Ceiling index	Title	Thickness (mm)
1	Panels 10	11
2	Panels 13	14
3	Panels 15	16.5

Note. Complied by the authors

Next, using the established encoding, a random process generates 170 different combinations of architectural features. For illustration purposes, Tables 7 and 8 display the random encodings and their corresponding parameters, respectively, for the first 20 and the last 10 scenarios. Table 7 presents the random combinations of architectural elements, using their corresponding indices designated in aforementioned tables. After that, the corresponding physical properties are shown in Table 8.

Based on these combinations, each scenario is modeled for energy calculation. For each case, the energy zones are applied to the functional areas in the building, as detailed in Table 9.

Table 7

Random Scenarios

Scenarios	Slab	Ceiling	External wall	Internal wall	External window	Primary door	Secondary door
1	3	3	3	2	5	1	2
2	3	3	2	3	1	2	1
3	1	1	2	1	4	1	1
4	2	2	3	3	6	1	1
5	2	2	2	3	4	1	1
6	1	1	2	3	5	3	1
7	3	3	3	1	3	1	1
8	1	1	2	2	5	3	2
9	1	1	3	3	6	2	2
10	3	3	4	2	5	2	2
11	2	2	4	3	1	2	1

Scenarios	Slab	Ceiling	External wall	Internal wall	External window	Primary door	Secondary door
12	1	1	2	3	1	1	2
13	2	2	2	1	5	3	2
14	1	1	4	2	3	2	1
15	3	3	2	1	3	3	2
16	2	2	4	2	1	2	2
17	3	3	4	1	2	2	1
18	3	3	2	1	2	3	2
19	1	1	3	2	3	1	2
20	3	3	3	3	1	3	1
…	…	…	…	…	…	…	…
161	3	3	2	3	6	2	2
162	2	2	3	3	2	1	1
163	2	2	4	2	3	2	1
164	1	1	2	2	3	2	2
165	2	2	4	3	4	2	1
166	1	1	4	3	1	3	2
167	1	1	3	3	4	2	1
168	3	3	4	2	6	2	1
169	2	2	3	3	3	3	1
170	2	2	2	2	3	2	2

Note. Complied by the authors

Table 8

Random Scenarios

Scenarios	Slab thickness (mm)	Ceiling thickness (mm)	Ex. Wall thickness (mm)	In. wall thickness (mm)	Ex. window area (m^2)	Ex. window layers	Prim. door area (m^2)	Sec. door area (m^2)
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
1	200	16.5	200	150	1.8	2	3.52	1.68
2	200	16.5	150	200	1.5	1	3.96	1.26
3	130	11	150	100	1.5	2	3.52	1.26
4	150	14	200	200	2.1	2	3.52	1.26
5	150	14	150	200	1.5	2	3.52	1.26

Scenarios	Slab thickness (mm)	Ceiling thickness (mm)	Ex. Wall thickness (mm)	In. wall thickness (mm)	Ex. window area (m^2)	Ex. window layers	Prim. door area (m^2)	Sec. door area (m^2)
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
6	130	11	150	200	1.8	2	4.4	1.26
7	200	16.5	200	100	2.1	1	3.52	1.26
8	130	11	150	150	1.8	2	4.4	1.68
9	130	11	200	200	2.1	2	3.96	1.68
10	200	16.5	250	150	1.8	2	3.96	1.68
11	150	14	250	200	1.5	1	3.96	1.26
12	130	11	150	200	1.5	1	3.52	1.68
13	150	14	150	100	1.8	2	4.4	1.68
14	130	11	250	150	2.1	1	3.96	1.26
15	200	16.5	150	100	2.1	1	4.4	1.68
16	150	14	250	150	1.5	1	3.96	1.68
17	200	16.5	250	100	1.8	1	3.96	1.26
18	200	16.5	150	100	1.8	1	4.4	1.68
19	130	11	200	150	2.1	1	3.52	1.68
20	200	16.5	200	200	1.5	1	4.4	1.26
…	…	…	…	…	…		…	…
161	200	16.5	150	200	2.1	2	3.96	1.68
162	150	14	200	200	1.8	1	3.52	1.26
163	150	14	250	150	2.1	1	3.96	1.26
164	130	11	150	150	2.1	1	3.96	1.68
165	150	14	250	200	1.5	2	3.96	1.26
166	130	11	250	200	1.5	1	4.4	1.68
167	130	11	200	200	1.5	2	3.96	1.26
168	200	16.5	250	150	2.1	2	3.96	1.26
169	150	14	200	200	2.1	1	4.4	1.26
170	150	14	150	150	2.1	1	3.96	1.68

Note. Complied by the authors

Table 9

Energy Zones

	Functional areas	Energy zones
Ground floor	Living room	Heating; cooling
	Bedroom	Heating; cooling
	Kitchen and dining room	Heating; cooling; ventilation
	Two toilets	DHW
	Staircase	Ventilation
First floor	Bedroom 1	Heating; cooling
	Bedroom 2	Heating; cooling
	Bedroom 3	Heating; cooling
	Living room	Heating; cooling
	Two toilets	DHW
	Laundry and drying spaces	Ventilation
	Staircase	Ventilation

Note. Complied by the authors

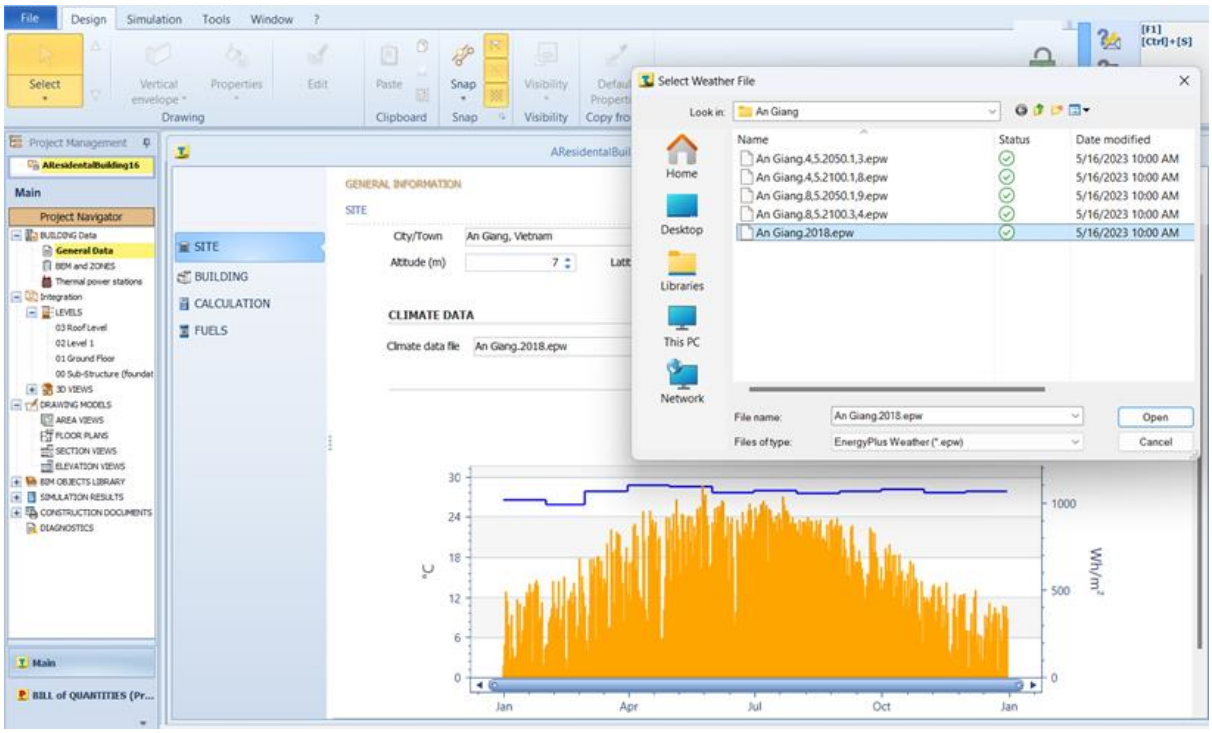
After defining the energy zones for the corresponding functional areas, the thermal power stations (HVAC and DHW systems) responsible for operating these zones are configured. In this study, two thermal power stations are assigned - one for each floor. It is important to note that the operation schedules for heating, cooling, and DHW systems are not standardized and can vary across buildings, even for those of the same type. Since the schedule is an intrinsic parameter in the simulation, its selection is independent of the core scientific principles underlying the proposed methodology. For this framework, a predefined schedule suitable for residential buildings is used to govern the operation of heating, cooling, and DHW systems. This schedule is provided by simulation software to ensure practical alignment with typical residential energy usage.

Next, the climate data is input into the software to define the external conditions for the building. The climate data used for the simulation corresponds to the most recent official records for An Giang province, recorded in 2018. This data is sourced from the Statistical Yearbook of Vietnam 2022, officially published by the General Statistics Office (GSO, 2022). Figure 10 illustrates the import of the 2018 climate profile into the simulation.

Thermal discontinuities at intersections of different BEM-objects within the building are addressed by modeling them as equivalent thermal bridges, with their properties calculated by the software. Figures 11 and 12 present the simulated model and the identified thermal bridge elements, respectively. Figure 13 provides an example of the calculation of the equivalent thermal properties for a thermal bridge element.

Figure 10

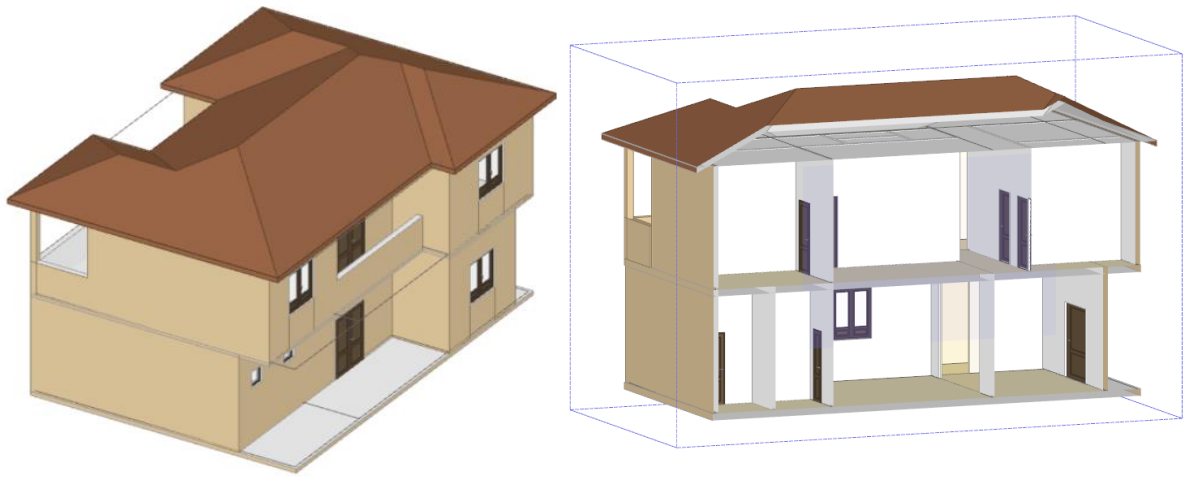
Climate Profile



Note. Complied by the authors

Figure 11

Views of Simulated Model

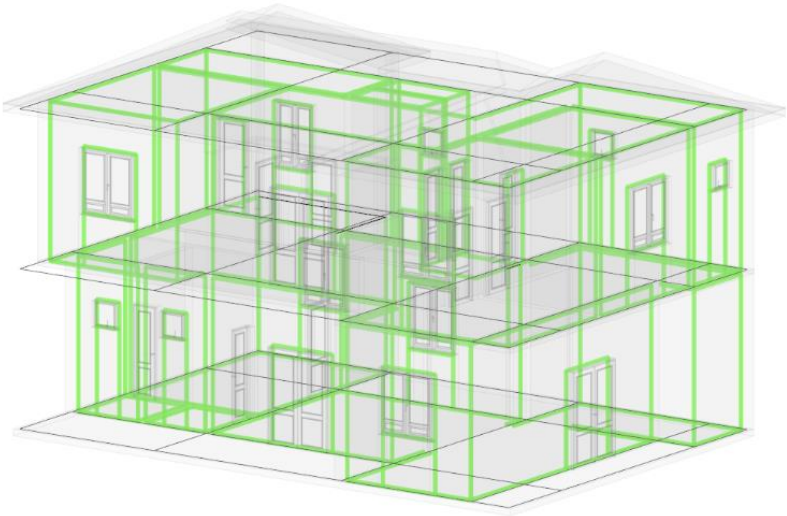


(a) General view of the simulated model (b) Sectional view of the simulated model

Note. Complied by the authors

Figure 12

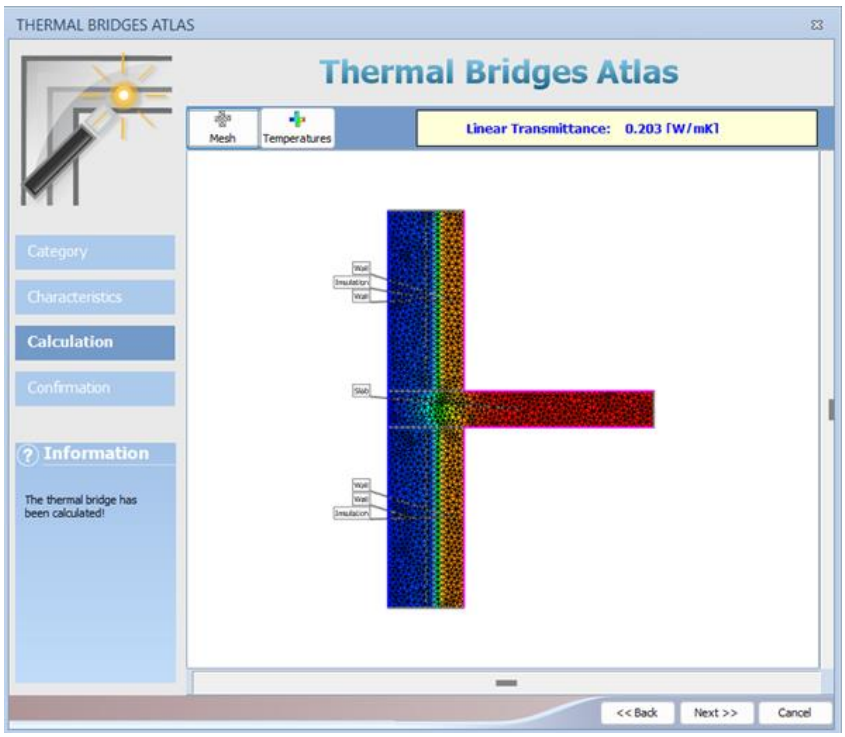
Detection of Thermal Bridges



Note. Complied by the authors

Figure 13

An Illustration of the Calculations for a Thermal Bridge

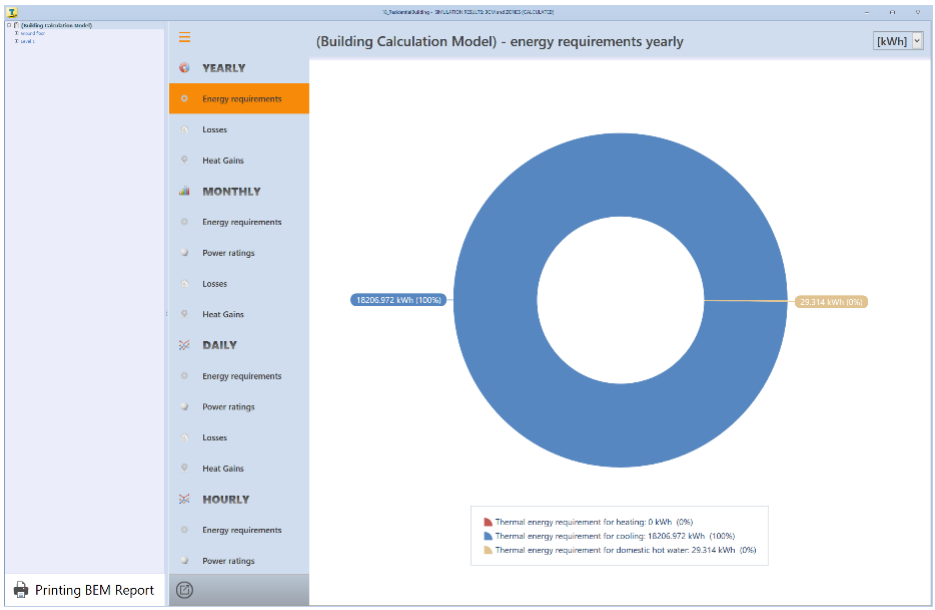


Note. Complied by the authors

With all BEM-objects and related settings in place, the energy simulation for the building is executed. The results from scenario 18 are presented as an example of the simulation outcomes. Figures 14 shows the annual energy requirements while Figure 15 illustrates the annual energy losses for the building. As shown in Figure 14, the energy requirements are predominantly driven by cooling, which is expected given the building's location in a tropical region with consistently high temperatures throughout the year.

Figure 14

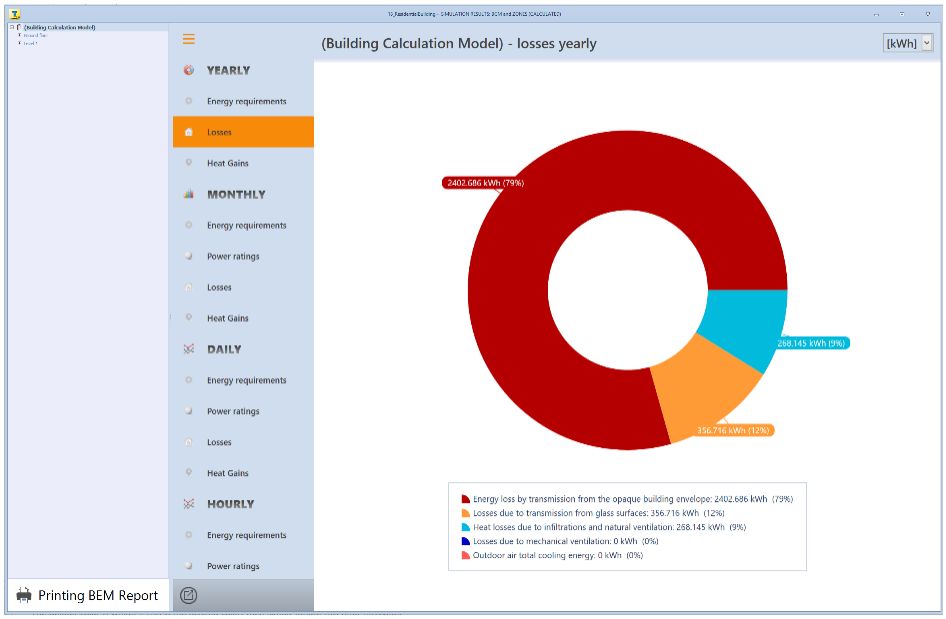
Energy Requirements for Scenario 18



Note. Complied by the authors

Figure 15

Energy Losses for Scenario 18



Note. Complied by the authors

3.2. Development of ML-based surrogate model

Upon completing the case-by-case simulations for all scenarios, a database is generated, as presented in Table 10. This study focuses on the building's energy requirements, so only this energy quantity is highlighted in the table. Using the generated database, an ANN-based surrogate model is developed through a fitting procedure to train the ML model. Hyperparameters are tuned using a trial-and-error approach, with the final configuration summarized in Table 11.

Table 10

Database

Scenarios	Slab thickness (mm)	Ceiling thickness (mm)	Ex. Wall thickness (mm)	In. wall thickness (mm)	Ex. window area (m^2)	Ex. window layers	Prim. door area (m^2)	Sec. door area (m^2)	Energy requirement (kWh)
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	y
1	200	16.5	200	150	1.8	2	3.52	1.68	16,712.009
2	200	16.5	150	200	1.5	1	3.96	1.26	17,810.587
3	130	11	150	100	1.5	2	3.52	1.26	17,948.541
4	150	14	200	200	2.1	2	3.52	1.26	16,454.101
5	150	14	150	200	1.5	2	3.52	1.26	17,330.402
6	130	11	150	200	1.8	2	4.4	1.26	17,497.339
7	200	16.5	200	100	2.1	1	3.52	1.26	17,593.642
8	130	11	150	150	1.8	2	4.4	1.68	17,730.638
9	130	11	200	200	2.1	2	3.96	1.68	16,831.731
10	200	16.5	250	150	1.8	2	3.96	1.68	16,070.029
11	150	14	250	200	1.5	1	3.96	1.26	16,726.302
12	130	11	150	200	1.5	1	3.52	1.68	18,117.465
13	150	14	150	100	1.8	2	4.4	1.68	17,772.514
14	130	11	250	150	2.1	1	3.96	1.26	17,069.679
15	200	16.5	150	100	2.1	1	4.4	1.68	18,238.838
16	150	14	250	150	1.5	1	3.96	1.68	16,920.829
17	200	16.5	250	100	1.8	1	3.96	1.26	16,975.081
18	200	16.5	150	100	1.8	1	4.4	1.68	18,236.286
19	130	11	200	150	2.1	1	3.52	1.68	17,694.376
20	200	16.5	200	200	1.5	1	4.4	1.26	17,192.023
…	…	…	…	…	…		…	…	
161	200	16.5	150	200	2.1	2	3.96	1.68	17554.039
162	150	14	200	200	1.8	1	3.52	1.26	17169.691
163	150	14	250	150	2.1	1	3.96	1.26	17338.310
164	130	11	150	150	2.1	1	3.96	1.68	16921.361
165	150	14	250	200	1.5	2	3.96	1.26	18345.431
166	130	11	250	200	1.5	1	4.4	1.68	16024.335
167	130	11	200	200	1.5	2	3.96	1.26	16873.065
168	200	16.5	250	150	2.1	2	3.96	1.26	16831.705
169	150	14	200	200	2.1	1	4.4	1.26	16070.269
170	150	14	150	150	2.1	1	3.96	1.68	17338.579

Note. Complied by the authors

Table 11

Energy Zones

Hyperparameters	Tuned results
Number of hidden layers	1
Number of nodes	100
Activation function	Rectified Linear Unit (ReLU) $\mathcal{A}(z) = \begin{cases} z\ , (z > 0) \\ 0\ , (z < 0) \end{cases}$
Gradient descent algorithm	Adamax [32] Learning rate: 0.002
Batch size	34

Note. Complied by the authors

To enhance the reliability of the predictive model, the ANN training process was performed twice, incorporating standardization and early stopping techniques. Each training used distinct splits of the dataset into training and testing sets. Specifically, in the first training, 136 samples were randomly selected for the training set, with the remaining 34 samples being used for testing. A similar procedure was repeated for the second training, ensuring that the 136-34 splits differed from those in the first. The training outcomes, evaluated using the selected metrics, are summarized in Table 12 and 13. The training history for both models is illustrated in Figures 16 and 17. It can be seen from Figures 16 and 17, the training of the ANN models through the tuned parameters exhibits to be steady, showing that the tuned neural network model is proper for the optimization scheme.

Additionally, the correlation between true and predicted values for the first and second ANN models is depicted in Figures 18 and 19, respectively. In those figures, not many irregular predictions are shown up.

The training results demonstrate the reliability of the tuned hyperparameters for the ANN model. Specifically, by evaluating the relative error of the Root Mean Square Error (*RMSE*) against the mean energy requirement, both models exhibit minimal error for both training and testing: 1.34% (training) and 1.66% (testing) for the first model, and 1.46% (training) and 1.49% (testing) for the second model. Additionally, the predicted results align closely with the true values, with $R_{training}$ = 0.9331 and $R_{testing}$ = 0.9094 for the first model, and $R_{training}$ = 0.9229 and $R_{testing}$ = 0.9181 for the second model. The outcomes from other metrics further confirm the accuracy and consistency of both models.

Table 12

ANN Model Performance Through Metric Evaluations

Metric	ANN model 1		ANN model 2	
	Training set	Testing set	Training set	Testing set
R	0.9331	0.9094	0.9229	0.9181
RMSE	231.4579	286.5039	251.3351	257.8793
MAE	166.2680	198.3833	175.3241	184.6495
Number of epochs	10000		10000	
Best epoch	9999		9997	

Note. Complied by the authors

Table 13

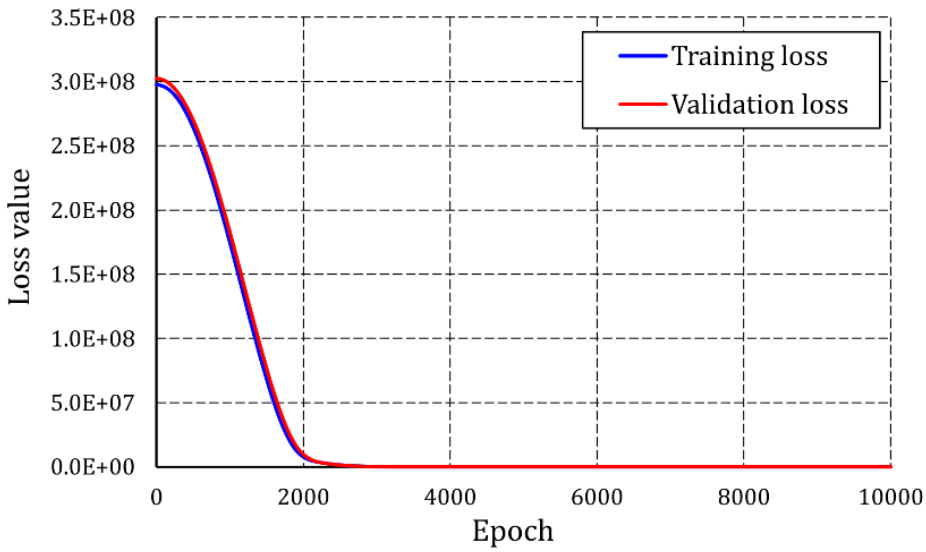
Relative Error for Energy Prediction by the ANN Models

Metric	ANN model 1		ANN model 2	
	Training set	Testing set	Training set	Testing set
$RMSE/\bar{\bar{E}}$	1.34%	1.66%	1.46%	1.49%

Note. Complied by the authors. $\bar{E}$ is the mean value of the energy requirements calculated over the scenarios

Figure 16

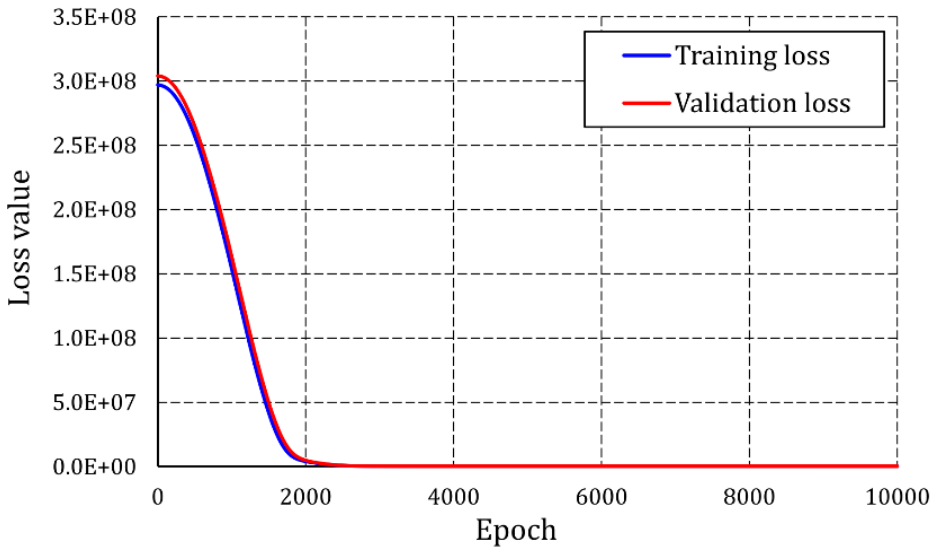
Training History of ANN Model 1



Note. Complied by the authors

Figure 17

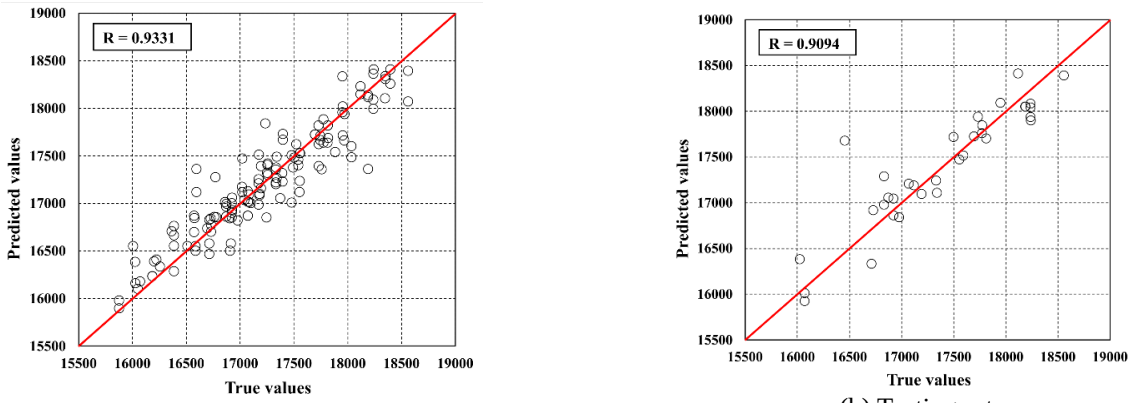
Training History of ANN Model 2



Note. Complied by the authors

Figure 18

Correlation between True and Predicted Values for ANN Model 1

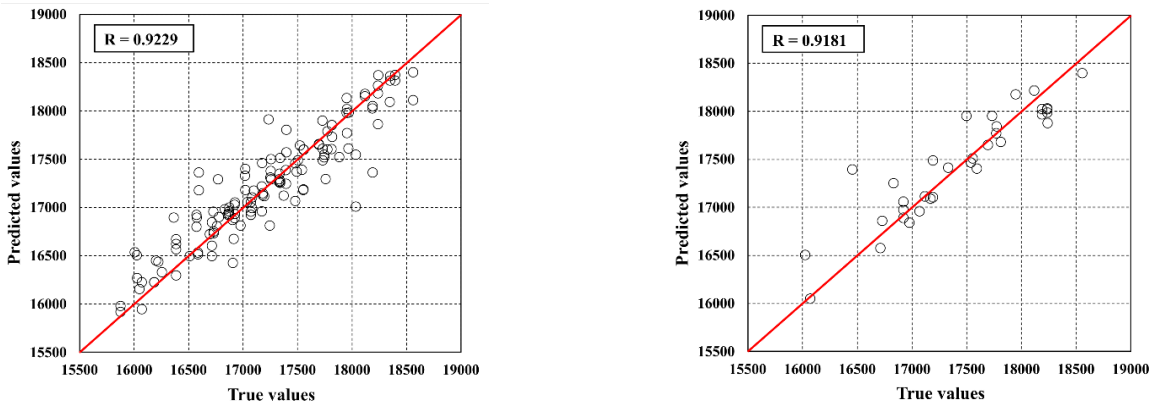


(a) Training set (b) Testing set

Note. Complied by the authors

Figure 19

Correlation between True and Predicted Values for ANN Model 2



(a) Training set (b) Testing set

Note. Complied by the authors

3.3. Optimization results

Within the scope of this project, an optimization problem is formulated with three variables to align with the size of the generated database. The selected variables include external wall thickness, internal wall thickness, and window area, while the remaining five architectural features - slab thickness, ceiling thickness, number of window layers, primary door area, and secondary door area - are held constant. The fixed parameters for these features are $x_1 = 150\ (mm)$, $x_2 = 11\ (mm)$, $x_6 = 2$ layers, $x_7 = 3.96\ (m^2)$, and $x_8 = 1.68\ (m^2)$, leading to the optimization vector being $\mathbf{x} = \{150, 11, x_3, x_4, x_5, 2, 3.96, 1.68\}$. Furthermore, architectural constraints suitable for residential buildings, following local design principles, are defined for the optimization process, as outlined in equations 9 to 12.

Wall thickness:

$$\frac{1}{16} H_{story} \leq x_3 \tag{9}$$

$$\frac{1}{32} H_{story} \leq x_4 \tag{10}$$

Total wall thickness by area:

$$\frac{x_3 L_3 + x_4 L_4}{A_{floor}} \leq 0.15 \tag{11}$$

Total window areas:

$$\frac{\sum x_5}{A_{floor}} \leq \frac{1}{4} \tag{12}$$

Using the defined constraints and the ANN-based surrogate model, the objective function for the energy optimization problem is formulated through a penalty-based approach, as shown in equations 13 to 14.

- Estimate energy requirement by ANN surrogate model:

$$\Phi(\mathbf{x}) = NN(\mathbf{x}) \tag{13}$$

- Apply the constraints using penalty approach:

$$P(\mathbf{x}) = r \begin{Bmatrix} max\left(0, \frac{1}{16}\frac{H_{story}}{x_3} - 1\right)^2 + max\left(0, \frac{1}{32}\frac{H_{story}}{x_4} - 1\right)^2 + \\ max\left(0, \frac{x_3 L_3 + x_4 L_4}{A_{floor}} - 0.15\right)^2 + max\left(0, \frac{\sum x_5}{A_{floor}} - \frac{1}{4}\right)^2 \end{Bmatrix} \tag{14}$$

where $r = 10^6, H_{story} = 3600\ (mm), L_3 = 99000\ (mm), L_4 = 45920\ (mm)$ and $A_{floor} = 139.88 \times 10^6 (mm^2)$

- Objective function:

$$F(\mathbf{x}) = \Phi(\mathbf{x}) + P(\mathbf{x}) \tag{15}$$

In the above formulations, H_{story} is the story height; L_3 is the total length of the external walls; L_4 is the total length of the internal walls; A_{floor} is the total floor area.

The optimization problem was solved over three independent runs to assess the stability of the results. The CPSO algorithm was initialized with 80 particles, using cognitive and social coefficients $c_1 = c_2 = 2.05$. The program terminates after 200 iterations, with the best solution extracted from the particle swarm. Table 14 summarizes the optimization outcomes, while Figures 20 to 22 illustrate the convergence trends for each run. The results in Table 14 demonstrate consistent performance across all three runs, with complete agreement in the optimal solutions for the design variables. In particular, all three runs give convergences of the designed variables to $[150, 11, \mathbf{223.7}, \mathbf{112.4}, \mathbf{1.9}, 2, 3.96, 1.68]$. Furthermore, the insignificance quantities of penalty terms, with only of 10-3 to 10-6, show that all the constraints of the optimization problem can be reached.

Table 14

Optimization Results

Resolved quantity	Optimized results
$\mathbf{x}_{optimized}$	$[150, 11, \mathbf{223.7}, \mathbf{112.4}, \mathbf{1.9}, 2, 3.96, 1.68]^{(run\ 1)}$
	$[150, 11, \mathbf{223.7}, \mathbf{112.4}, \mathbf{1.9}, 2, 3.96, 1.68]^{(run\ 2)}$
	$[150, 11, \mathbf{223.7}, \mathbf{112.4}, \mathbf{1.9}, 2, 3.96, 1.68]^{(run\ 3)}$
$F(\mathbf{x}_{optimized})$	$16319.609^{(run\ 1)}$
	$16319.607^{(run\ 2)}$
	$16319.607^{(run\ 3)}$
$\Phi(\mathbf{x}_{optimized})$	$14240.621^{(run\ 1)}$
	$14240.685^{(run\ 2)}$
	$14240.688^{(run\ 3)}$
$P(\mathbf{x}_{optimized})$	$2078.898^{(run\ 1)}$
	$2078.923^{(run\ 2)}$
	$2078.920^{(run\ 3)}$
$p_i(\mathbf{x}_{optimized})$	$[3.57 \times 10^{-5}, 1.11 \times 10^{-6}, 2.04 \times 10^{-3}, 0.00]^{(run\ 1)}$
	$[3.59 \times 10^{-5}, 1.17 \times 10^{-6}, 2.04 \times 10^{-3}, 0.00]^{(run\ 2)}$
	$[3.60 \times 10^{-5}, 1.15 \times 10^{-6}, 2.04 \times 10^{-3}, 0.00]^{(run\ 1)}$

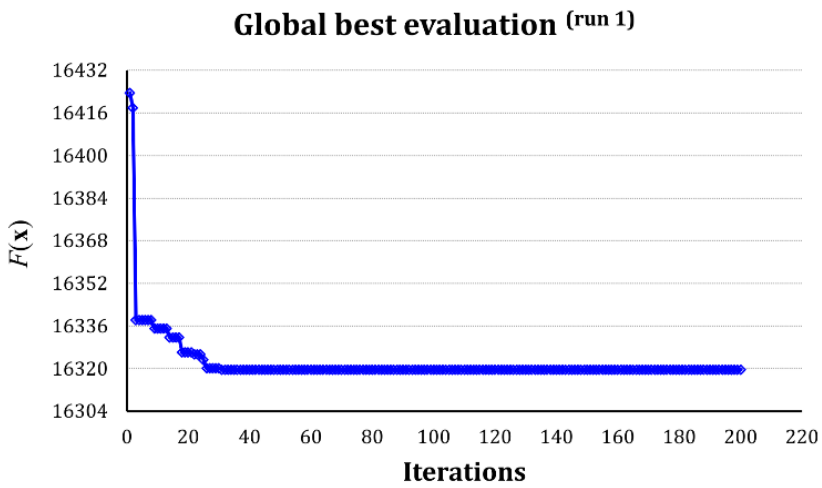
$$\{p_i(\mathbf{x}_{optimized})\} = \left[max\left(0, \frac{1}{16}\frac{H_{story}}{x_3} - 1\right)^2, max\left(0, \frac{1}{32}\frac{H_{story}}{x_4} - 1\right)^2, max\left(0, \frac{x_3 L_3 + x_4 L_4}{A_{floor}} - 0.15\right)^2, max\left(0, \frac{\sum x_5}{A_{floor}} - \frac{1}{4}\right)^2\right]$$

Note. Complied by the authors

The fitness function evaluations for the optimized solution are nearly identical across runs, with values slightly above 16,319.6kWh. A similar trend is observed in the evaluations of the surrogate model, which consistently yields results around 14,240.6kWh, confirming the effectiveness of the optimization procedure. Additionally, the penalty function results across the three runs fluctuate around 2,078.9kWh, which is negligible given the penalty coefficient of $r = 10^6$. All penalty function components $\{p_i(\mathbf{x}_{optimized})\}$ evaluated at the optimal solution approach zero, confirming the rationality of the optimization process.

Figure 20

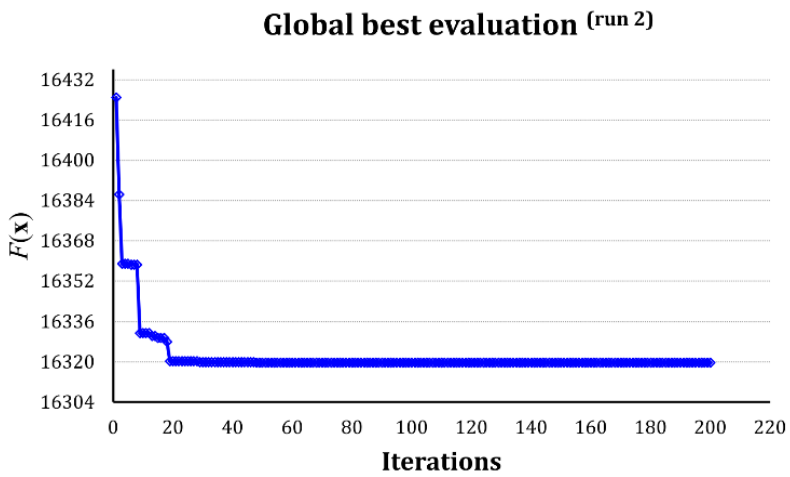
Convergence of the Optimization Process (Run 1)



Note. Complied by the authors

Figure 21

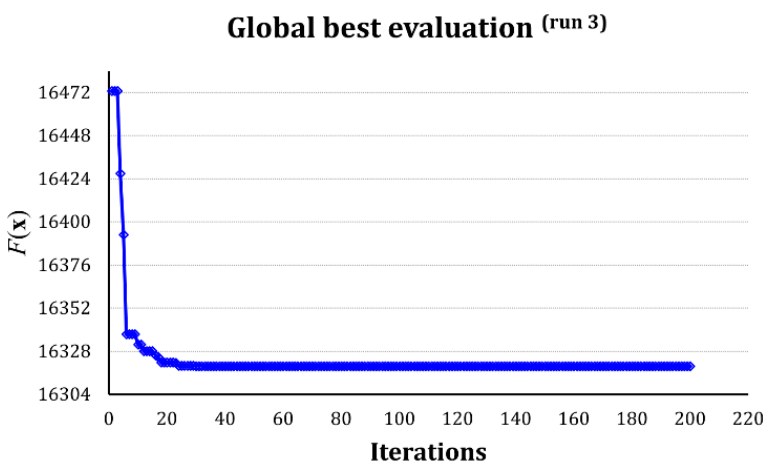
Convergence of the Optimization Process (Run 2)



Note. Complied by the authors

Figure 22

Convergence of the Optimization Process (Run 3)



Note. Complied by the authors

As shown in Figures 20 to 22, the optimization converges stably in all three runs, reaching the minimum fitness value after approximately 40 iterations.

4. Discussions and future research directions

Our work at this stage is focused on an application-oriented study which is to provide a methodology for developing energy-efficient architectural designs. To this end, a widely used framework is inherited from the literature to from a practical application in the context of constructions in Vietnam with environment's impacts considered.

To demonstrate its practical application, a case study focusing on the design of a residential building was conducted, incorporating real-world data, such as climate conditions and local architectural principles, to enhance the practicality of the approach. Based on the results, several key discussions and potential future directions for research are outlined.

First, within the scope of this project, a modest dataset of 170 scenarios was generated, which limited the complexity of the optimization problem. Specifically, only three architectural features were selected as optimization variables to ensure the soundness of the framework. However, this limitation does not compromise the scientific validity of the methodology. As the database size increases, the ML-based surrogate model can be extended to accommodate a broader set of variables with diverse ranges, allowing for more sophisticated optimization strategies while maintaining accuracy and reliability.

Second, the case study results highlight the promising potential of the proposed methodology for practical building design. Several research directions are envisioned to advance this preliminary work toward real-world applications. A future study will focus on energy modeling based on experimental validation, with parameter calibration to enhance the precision and reliability of energy quantification, ensuring the generation of a robust database. Additionally, advanced ML techniques capable of handling large datasets will be employed to further develop the surrogate model. This would allow for more complex optimization problems, incorporating a wider range of design variables to reflect realistic architectural scenarios. Ultimately, this approach offers a viable path for improving building energy performance by providing architects with effective strategies and actionable design insights.

5. Conclusion

This study inherited a widely used framework within the field to develop a application-oriented methodology to identify optimal solutions for building energy efficiency, with a focus on architectural design in the context of Vietnam's construction on environment. The significance of this study lies on the employment of an applicable methodology from the literature into a practical design process which can be applied in Vietnam. The methodology was demonstrated through a case study of a residential building located in An Giang province, Vietnam, incorporating architectural constraints within the optimization framework.

An ANN-based surrogate model was employed to map design variables directly to the building's energy requirements, significantly reducing the computational cost of energy simulations. Once the ANN model's reliability was confirmed, it was integrated into the objective function for the optimization process. The results of the optimization, along with the relative importance analysis based on the surrogate model, were presented.

The case study results highlight the potential of the proposed approach for improving building energy performance while accounting for architectural factors. Several insights are

drawn from the findings: (i) the complexity of the optimization problem was limited to align with the available database size; (ii) to transition this preliminary research into practical applications, it is essential to tailor the energy simulation model for accurate prediction, ensuring experimental validation and parameter calibration. These steps will strengthen the framework's applicability in real-world architectural design; and (iii) the application of the presented procedure does not limit to the choice of machine learning model, by which based on the nature of the problem to be solved, a suitable can be tuned.

SCIENTIFIC CONTRIBUTION

The manuscript clearly identifies the necessity; the manuscript proposes an application-oriented study; the manuscript opens new directions for further research.

AUTHOR CONTRIBUTIONS

CRediT: **Le Quang Huy**: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Writing - Original Draft, Writing - Review & Editing, Supervision; **Hoang Quynh Anh**: Conceptualization, Architecture drawing, Resources, Data Curation, Writing - Original Draft, Visualization.

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NO CONFLICT OF INTEREST STATEMENT

All authors declare that they have no conflict of interest.

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